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Green's-function diagrammatic technique for complicated level systems*

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A Green's function theory applicable to magnetic systems with complicated levels for each ion is developed using the standard basis operators. A diagrammatic technique is employed. This provides a self-consistent approach to the rare-earth and actinide systems where the crystal field is of the same order as the exchange interaction. A Wick-like theorem is shown first followed by a discussion of the diagrammatic representation. Guided by a priority rule we consider all possible vertices. Rules for construction and evaluation of diagrams are given and summation of diagrams is discussed.

I. INTRODUCTION

In the field of magnetism, as in most of the areas of solid state physics, the many-body problem plays the central role. Because the exact solution cannot generally be found except for a few low-dimensional systems, the effective field approximation has been commonly employed to reduce the problem. But because of the extreme crudeness of the approximation the results of the calculation can be deemed only qualitatively correct. At low temperatures spin-wave theory can be applied to certain simple systems; the validity is, however, restricted to temperatures much less than the transition temperature. The attempt to cover the behavior of the system for the whole temperature range and to give a reasonable description even near the phase transition has been made with the aid of a Green's-function technique. Successful results have been obtained for simple systems such as the isotropic Heisenberg ferromagnet.[1,2] In reality, however, there exist, in almost all the magnetic systems, crystal-field anisotropy, dipolar and anisotropic exchange interactions, and in some cases higher-order multipole-multipole interactions. The energy levels of each individual ion in the effective-field approximation can become so complicated that even a spin-wave theory would be difficult to construct, let alone the more sophisticated Green's-function theory. This is particularly true in rare-earth and actinide systems, because the crystal-field interaction is of the same order of magnitude as the exchange interaction. Even in some transition-element compounds complications arise when the orbital momentum is not totally quenched.

It is the purpose of this paper to develop a Green's-function formalism applicable to the above-mentioned complicated-level many-body systems. Recently a pseudospin Green's-function formalism has been used to discuss the behavior of the simple two-level induced-moment systems.[3] For systems of more energy levels for each ion the formalism becomes formidably complicated. The spin-1 Heisenberg ferromagnet with a single-ion anisotropy giving rise to an easy axis has also been a subject of many papers. Approximate Green's functions have been obtained both by the equation-of-motion technique[4,5] and by the diagrammatic expansion.[6] Inconsistencies arising from the decoupling approximation in the equation-of-motion technique were noted by Murao and Matsubara and remains in the subsequent treatments by other authors using similar methods. While the alternative diagrammatic-expansion technique recently developed for spin operators[2] can be used to ensure the consistency in the calculation, it becomes rather complicated even for a system as simple as this. There is a slim hope therefore in adapting the method to a more complicated system.

We shall adopt a diagrammatic approach, not only because it enables one to recognize the physical processes and the approximations involved in the calculation, but more importantly because it can ensure the self-consistency of the calculation. The diagrammatic technique developed for spin operators by Vaks, Larkin, and Pikin[2] (VLP) is not practical in dealing with systems other than the simplest ones like the isotropic Heisenberg or Ising systems. A much more general and efficient diagrammatic technique is needed. This can be developed by using the standard basis operators originally introduced by Hubbard[7] in the atomic representation formulation, and later used by Haley and Erdös[5] in the discussion of a spin one anisotropic Heisenberg ferromagnetic system. The idea is that the effective single-ion Hamiltonian can be solved generally without much difficulty. Then using the eigenstates of the effective field Hamiltonian as the basis one can reformulate the problem by projecting the total Hamiltonian onto this manifold. To describe the manifold of states and the transitions of an ion from one state to another, operators which annihilate an ion in a state and simultaneously create the ion in another state (or the same state) are defined. The operators are called standard basis operators in Ref. 5. Any spin operator or product of spin operators of the same site can be expressed as a linear combination of these operators and the Hamiltonian is therefore recast in the new language. In its new form the Hamiltonian is greatly simplified. Unlike the original Hamiltonian in spin operators the Hamiltonian expressed in the standard basis operators contains only operators of the first order for each site. A Wick-like theorem can be formulated for the standard operators taking all the single-ion terms as the unperturbed part of the Hamiltonian. This makes possible a diagrammatic analysis of Green's functions on the basis of the effective field eigenstates. Very recently Westwanski and Pawlikowski[8] have reported briefly a Wick-like reduction theorem using the standard basis operators. No detailed discussion of the diagrammatic technique has been given.

We shall in Sec. II review briefly the properties of the standard basis operators and derive the Wick-like theorem in a similar procedure as used by VLP for the spin operators. A general diagrammatic representation is then developed and all possible interaction vertices are shown. In Sec. III rules for construction of diagrams for a Green's function and the evaluation of each diagram are discussed. Summation of simple chain diagrams for certain special systems such as the induced-moment system $PrTl_3$ is shown in Sec. IV as a simple illustration of the method. A more complete exposure of the techniques developed in this paper will be given in a subsequent publication in which a Heisenberg ferromagnet with a single-ion anisotropy will be discussed in a fully self-consistent way.

II. STANDARD BASIS OPERATORS AND THE WICK-LIKE THEOREM

Following Refs. 5 and 7, we assume for each ion a set of energy levels obtained, say, by solving the single-ion effective field Hamiltonian, and define the standard basis operators as

$$L_{\alpha\beta} = |\alpha\rangle\langle\beta| \ . \tag{1}$$

The operator $L_{\alpha\beta}$, therefore, will transfer an ion from the state $|\beta\rangle$ to the state $|\alpha\rangle$. The multiplication rules for the operators are evident;

$$L_{\alpha\alpha'} L_{\beta\beta'} = \delta_{\alpha'\beta} L_{\alpha\beta'} \ . \tag{2}$$

Operators of different sites obviously commute. The commutation relations of operators of the same site are

$$[L_{\alpha\alpha'}, L_{\beta\beta'}] = \delta_{\alpha'\beta} L_{\alpha\beta'} - \delta_{\alpha\beta'} L_{\beta\alpha'} \ . \tag{3}$$

The operator $L_{\alpha\alpha}$ measures the probability that the energy level α is occupied. Its ensemble average will be denoted by D_α. Since the sum of $L_{\alpha\alpha}$ over all the energy levels is equal to an identity operator we have the sum rule

$$\sum_\alpha D_\alpha = 1 \ . \tag{4}$$

To formulate the problem in the language of standard basis operators, we first note that any operator can be expressed as a linear combination of the standard basis operator by the formula

$$O = \sum_{\alpha,\beta} \langle \alpha | O | \beta \rangle L_{\alpha\beta} \ . \tag{5}$$

Choosing the eigenstates of the single-ion Hamiltonian as the basis states, the total Hamiltonian takes the general form

$$\mathcal{H} = \sum \epsilon_\alpha L^n_{\alpha\alpha} + \frac{1}{2} \sum V^{n,m}_{\alpha\alpha',\beta\beta'} L^n_{\alpha\alpha'} L^m_{\beta\beta'} \ , \tag{6}$$

where we have used superscripts as site indices to label the operators, and each summation is over all corresponding indices. ϵ_α is the eigenenergy of the effective field Hamiltonian and $V^{n,m}_{\alpha\alpha',\beta\beta'}$ represents a two-body interaction vertex, $\langle \alpha\beta | V | \alpha'\beta' \rangle$. Writing $\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{\text{int}}$, we identify the single-ion part of the Hamiltonian with the unperturbed Hamiltonian $\mathcal{H}_0$, and the two-ion interaction potential with $\mathcal{H}_{\text{int}}$.

We now consider the temperature-dependent Green's function

$$G_{\alpha\beta,\gamma\delta}(r_1, \tau_1; r_2, \tau_2) \equiv \langle T_\tau(\tilde{L}_{\alpha\beta}(r_1, \tau_1) \tilde{L}_{\gamma\delta}(r_2, \tau_2)) \rangle \ , \tag{7}$$

where the T_τ operator arranges the order of the operators in decreasing order of τ from left to right. The angular brackets denote a canonical ensemble average with respect to $\mathcal{H}$ and the operators in the Green's function are in the Heisenberg representation. It is a standard procedure to expand the Green's function in the form

$$\langle T_\tau(\tilde{L}_{\alpha\beta}(r_1, \tau_1) \tilde{L}_{\gamma\delta}(r_2, \tau_2)) \rangle = \langle T_\tau(L_{\alpha\beta}(r_1, \tau_1) L_{\gamma\delta}(r_2, \tau_2) \mathcal{S}(\beta)) \rangle_0 / \langle \mathcal{S}(\beta) \rangle_0 \ , \tag{8}$$

where the S matrix $\mathcal{S}(\beta)$ is defined as

$$\mathcal{S}(\beta) = \sum_{n=0}^\infty \frac{(-1)^n}{n!} \int_0^\beta \cdots \int_0^\beta \times T_\tau(\mathcal{H}_{\text{int}}(\tau_1) \mathcal{H}_{\text{int}}(\tau_2) \cdots) \, d\tau_1 \ldots d\tau_n \ , \tag{9}$$

or in a short-hand form as

$$\mathcal{S}(\beta) = T_\tau \exp\left(- \int_0^\beta \mathcal{H}_{\text{int}}(\tau) \, d\tau \right) \ , \tag{10}$$

and $\beta = 1/k_B T$. On the right-hand side of Eq. (8) the canonical average is taken with respect to the unperturbed Hamiltonian $\mathcal{H}_0$ and the operators are in the interaction representation. A complete discussion of the temperature-dependent Green's function is given in Ref. 9.

The problem of evaluation of the Green's function is now reduced to the evaluation of the average of the τ ordered product of operators over the ensemble of noninteracting ions. In the fermion or boson problems, Wick's theorem and diagrammatic representation are used to facilitate the evaluation of the average of the T products. A Wick-like theorem can also be found for the standard basis operators. We shall follow the procedure used by VLP, and later described in more detail by Izyumov and Kasan-Ogly,[10] to construct the theorem.

We first note the τ dependence of the standard basis operators in the interaction representation. Recalling that

$$\mathcal{H}_0 = \sum \epsilon_\alpha L_{\alpha\alpha} , \tag{11}$$

and the relation of the interaction representation to the Schrödinger representation,

$$L_{\alpha\beta}(\tau) = e^{\mathcal{H}_0\tau} L_{\alpha\beta} e^{-\mathcal{H}_0\tau} , \tag{12}$$

we obtain

$$L_{\alpha\beta}(\tau) = e^{(\epsilon_\alpha - \epsilon_\beta)\tau} L_{\alpha\beta} . \tag{13}$$

Now consider the thermal average of the T product $\langle T_\tau(O_1(\tau_1)\ldots O_m(\tau_m)L_{\alpha\beta}(\tau)O_{m+1}(\tau_{m+1})\ldots O_n(\tau_n))\rangle_0$, where each O represents a standard basis operator $L_{\gamma\delta}$. In the product, all the operators are on the same lattice site; the thermal average of a product of operators of different sites factors because the ensemble consists of noninteracting ions. It is clear that any rearrangement of the operators within the T product is permissible. Without loss of generality we assume that the operators are already ordered in τ. So we have

$$\langle T_\tau(O_1(\tau_1)\cdots L_{\alpha\beta}(\tau)\cdots O_n(\tau_n))\rangle_0 = \langle O_1(\tau_1)\cdots L_{\alpha\beta}(\tau)\cdots O_n(\tau_n)\rangle_0 , \tag{14}$$

which is by definition $(1/Z_0)\,\mathrm{Tr}e^{-\beta\mathcal{H}_0}(O_1(\tau_1)\cdots L_{\alpha\beta}(\tau)\cdots O_n(\tau_n))$, where $Z_0 = \mathrm{Tr}e^{-\beta\mathcal{H}_0}$. We now move the operator $L_{\alpha\beta}(\tau)$ to the left with the aid of the identity

$$O_i(\tau_i)L_{\alpha\beta}(\tau) = L_{\alpha\beta}(\tau)O_i(\tau_i) + [O_i(\tau_i),\, L_{\alpha\beta}(\tau)] . \tag{15}$$

The process of moving $L_{\alpha\beta}(\tau)$ over the operator to its left generates an extra product which consists of a commutator of $L_{\alpha\beta}(\tau)$ with that operator and the other $n-1$ operators. This process is repeated until $L_{\alpha\beta}(\tau)$ is moved to the left of O_1, next to the exponential factor. Then we use the identity

$$e^{-\beta\mathcal{H}_0} L_{\alpha\beta}(\tau) = e^{-\beta(\epsilon_\alpha - \epsilon_\beta)} L_{\alpha\beta}(\tau) e^{-\beta\mathcal{H}_0} \tag{16}$$

to move $L_{\alpha\beta}$ further over the exponential factor and use the cyclic invariance of trace to bring it around to the rear. Resuming the process of moving $L_{\alpha\beta}$ to the left step-by-step, finally we recover the original product multiplied by a factor $e^{-\beta(\epsilon_\alpha - \epsilon_\beta)}$. There are also n other products generated, each of which contains $n-1$ operators and a commutator $[O_i(\tau_i),\, L_{\alpha\beta}(\tau)]$, and carries a factor $e^{-\beta(\epsilon_\alpha - \epsilon_\beta)}$ if $\tau_i < \tau$. The commutator can be further written as

$$e^{(\epsilon_\alpha - \epsilon_\beta)(\tau - \tau_i)}[O_i(\tau_i),\, L_{\alpha\beta}(\tau_i)]$$

or

$$e^{-(\epsilon_\alpha - \epsilon_\beta)(\tau_i - \tau)}[O_i,\, L_{\alpha\beta}]_{\tau_i} .$$

Thus, if we define a noninteracting Green's function $G^0_{\alpha\beta}(\tau_i - \tau)$ as

$$G^0_{\alpha\beta}(\tau_i - \tau) = \begin{cases} e^{-(\epsilon_\alpha - \epsilon_\beta)(\tau_i - \tau)}[1 + n(\epsilon_\alpha - \epsilon_\beta)] , & \tau_i > \tau \\ e^{-(\epsilon_\alpha - \epsilon_\beta)(\tau_i - \tau)} n(\epsilon_\alpha - \epsilon_\beta) , & \tau_i < \tau \end{cases} \tag{17}$$

where $n(\epsilon) = (e^{\beta\epsilon} - 1)^{-1}$. We obtain from the above manipulation

$$\begin{aligned}\langle T_\tau(O_1(\tau)\cdots L_{\alpha\beta}(\tau)\cdots O_n(\tau_n))\rangle_0 &= G^0_{\alpha\beta}(\tau_1 - \tau)\langle T_\tau([O_1,\, L_{\alpha\beta}]_{\tau_1}\cdots O_n(\tau_n))\rangle_0 \\ &\quad + G^0_{\alpha\beta}(\tau_2 - \tau)\langle T_\tau(O_1(\tau_1) \\ &\quad \times [O_2,\, L_{\alpha\beta}]_{\tau_2}\cdots O_n(\tau_n))\rangle_0 + \cdots \\ &\quad + G^0_{\alpha\beta}(\tau_n - \tau)\langle T_\tau(O_1(\tau_1)\cdots [O_n,\, L_{\alpha\beta}]_{\tau_n})\rangle_0 .\end{aligned} \tag{18}$$

This can be stated literally as matching one of the operators in the product with all the other operators. Each matching produces a commutator and a Green's function. Noting that each commutator gives only one operator or a linear combination of operators, we have obtained a reduction theorem which reduces the average of a T product of $n+1$ operators to a sum of terms each of which is a product of a Green's function and an average of a T product of n operators. Using the reduction theorem repeatedly for the other operators we obtain the Wick-like theorem for the standard basis operators. If we call $L_{\alpha\beta}$ a diagonal operator when $\alpha = \beta$, the thermal average of a T product can then be expressed as a sum of products of Green's functions and thermal averages of diagonal operators.

Instead of writing down the complete form of the Wick-like theorem, it is more instructive to illustrate the theorem with a few simple examples. While doing this we shall also introduce the diagrammatic representation.

As a first example, we consider the average,

$$\langle T_\tau(L^1_{12}(\tau_1)L^2_{21}(\tau_2)L^3_{11}(\tau_3))\rangle_0 .$$

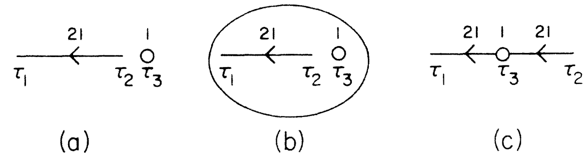

FIG. 1. Diagrams for $\langle T_\tau L^1_{12}(\tau_1)L^2_{21}(\tau_2)L^3_{11}(\tau_3)\rangle_0$.

It is to be recalled that the superscript of each operator denotes the lattice site, and the subscripts enumerate the energy levels. Using the reduction scheme derived above and starting the process with $L^2_{21}(\tau_2)$, we obtain

$$\delta_{12}G^0_{21}(\tau_1-\tau_2)\langle(L^1_{11}-L^1_{22})L^3_{11}\rangle_0-\delta_{23}G^0_{21}(\tau_3-\tau_2)\times\delta_{13}G^0_{21}(\tau_1-\tau_3)\langle L^1_{11}-L^1_{22}\rangle_0\,,$$

which is,

$$\delta_{12}G^0_{21}(\tau_1-\tau_2)[D_1D_{12}+\delta_{13}(D_1-D_1D_{12})]-\delta_{23}G^0_{21}(\tau_3-\tau_2)\delta_{13}G^0_{21}(\tau_1-\tau_3)D_{12}\,, \tag{19}$$

where we have introduced the notation $D_{\alpha\beta}\equiv D_\alpha-D_\beta$ and $D_\alpha\equiv\langle L_{\alpha\alpha}\rangle_0$ as defined previously. The three terms obtained can be represented by three diagrams. We use a solid line drawn from τ_2 to τ_1 to represent $G^0_{21}(\tau_1-\tau_2)$. The line carries an arrow pointing from τ_2 to τ_1 and is labeled with 21. We use a small circle to represent a diagonal vertex $L_{\alpha\alpha}$ and label the vertex with α. The three diagrams are shown in Fig. 1. The second diagram requires an explanation. In this diagram not only site 1 is required to be the same as site 2, as the Green's function would vanish otherwise, but site 3 is also restricted to be the same as site 1 and 2. To indicate such a restriction we have used an oval to embrace the two parts of the diagram. The occurrence of this type of diagram merits a comment. As shown in the above calculation, the second diagram is created because the thermal average $\langle(L^1_{11}-L^1_{22})L^3_{11}\rangle_0$ depends on whether site 3 is the same as site 1. When they are of the same site, the average is D_1 instead of the product of the separate averages $D_{12}D_1$, reminiscent of the fact that $\langle(S^z)^2\rangle_0$ is not equal to $\langle S^z\rangle_0^2$ in the spin-operator formulation. In the first diagram, lattice site 3 is not restricted and the diagram includes the case that site 3 is same as site 1. The weight factor is nonetheless always $D_{12}D_1$. The second diagram therefore represents a correction to the weight factor and the sum of the two diagrams correctly reproduces the weight factor $\langle(L^1_{11}-L^1_{22})L^3_{11}\rangle_0$. The fact that the weight factor for the second diagram is not simply the product of weight factors associated with the parts of the diagram when they are separate precludes the use of the original form of Dyson's equation in summing the diagrams, as will be discussed later. The third diagram represents a term obtained actually by applying the reduction theorem twice. The process is to match first $L^2_{21}(\tau_2)$ with $L^3_{11}(\tau_3)$ and then to match the resultant operator $L^2_{21}(\tau_3)$ with $L^1_{12}(\tau_1)$. Physically it shows a scattering of the excitation against a diagonal vertex.

The second simple example which will show some other important features of the scheme is the calculation of the average

$$\langle T_\tau(L^1_{12}(\tau_1)L^2_{21}(\tau_2)L^3_{32}(\tau_3)L^4_{23}(\tau_4))\rangle_0\,.$$

If we start the reduction process with $L^2_{21}(\tau_2)$, and continue it with $L^3_{32}(\tau_3)$ or $L^3_{31}(\tau_3)$ which is generated in the first matching, we obtain,

$$\begin{aligned}\langle T_\tau(L^1_{12}(\tau_1)L^2_{21}(\tau_2)L^3_{32}(\tau_3)L^4_{23}(\tau_4))\rangle_0 ={}& \delta_{12}G^0_{21}(\tau_1-\tau_2)\delta_{34}G^0_{32}(\tau_4-\tau_3)[D_{12}D_{23}-\delta_{14}(D_2+D_{12}D_{23})]\\ &+\delta_{12}G^0_{21}(\tau_1-\tau_2)\delta_{31}G^0_{32}(\tau_1-\tau_3)\delta_{14}G^0_{32}(\tau_4-\tau_1)D_{23}\\ &-\delta_{23}G^0_{21}(\tau_3-\tau_2)\delta_{31}G^0_{31}(\tau_1-\tau_3)\delta_{14}G^0_{32}(\tau_4-\tau_1)D_{23}\\ &+\delta_{23}G^0_{21}(\tau_3-\tau_2)\delta_{34}G^0_{31}(\tau_4-\tau_3)\delta_{14}G^0_{21}(\tau_1-\tau_4)D_{12}\,.\end{aligned} \tag{20}$$

The corresponding diagrams for the above terms are shown in Fig. 2. Two types of new diagrams emerge. Diagram (c) has three Green's-function lines meeting at one point. The diagram reflects the kinematics of the standard basis operator. Here a correction is made to enforce the restriction that only one of the energy levels of an ion can be occupied. Mathematically it arises from a scattering of the excitation from level 2 to level 3 against the diagonal vertex produced at the termination of the excitation from level 1 to level 2. Diagrams (d) and (e) are simple to interpret. They describe successive excitations among the three levels.

The set of diagrams obtained above is, however, not unique. If we start out the process of reduction with $L^3_{32}(\tau_3)$, and continue it with $L^2_{21}(\tau_2)$ or $L^2_{31}(\tau_2)$, we obtain

$$\begin{aligned}&\langle T_\tau(L^1_{12}(\tau_1)L^2_{21}(\tau_2)L^3_{32}(\tau_3)L^4_{23}(\tau_4))\rangle_0\\ &=\delta_{12}G^0_{21}(\tau_1-\tau_2)\delta_{34}G^0_{32}(\tau_4-\tau_3)[D_{12}D_{23}-\delta_{14}(D_2+D_{12}D_{23})]+\delta_{34}G^0_{32}(\tau_4-\tau_3)\delta_{24}G^0_{21}(\tau_4-\tau_2)\delta_{14}G^0_{21}(\tau_1-\tau_4)D_{12}\\ &\quad-\delta_{32}G^0_{32}(\tau_2-\tau_3)\delta_{24}G^0_{31}(\tau_4-\tau_2)\delta_{14}G^0_{21}(\tau_4-\tau_1)D_{12}+\delta_{32}G^0_{32}(\tau_2-\tau_3)\delta_{21}G^0_{31}(\tau_1-\tau_2)\delta_{14}G^0_{32}(\tau_4-\tau_1)D_{23}\,.\end{aligned} \tag{21}$$

The corresponding diagrams are shown in Fig. 3. Except for the first two diagrams, the two sets of diagrams are quite different. It is obvious that the reduction scheme we have found and used does not give results in a unique form. This, however, does not constitute a setback for the scheme. In fact, the lack of uniqueness of the representation also prevails in the formalism using spin operators.

To unify as well as simplify the diagram representation we set up a priority rule as an auxiliary guidance in the reduction procedures. The rule consists of two parts.

(a) All diagonal operators like $L_{\alpha\alpha}$ and operators $L_{\alpha\delta}$ with $\alpha < \delta$ remain stationary in the reduction process. They may be called passive operators.

(b) The active operators $L_{\alpha\delta}$ with $\alpha > \delta$ match with all the operators in the product in a given order depending on δ and α. Priority of matching is given to the operator with the smallest δ. If there is more than one operator with the smallest δ, priority is given to the one with the smallest α among them.

According to this priority rule we choose the set of diagrams shown in Fig. 2 to represent the average of the product $\langle T_\tau(L_{12}^1(\tau_1)L_{21}^2(\tau_2)L_{32}^3(\tau_3) \times L_{23}^4(\tau_4))\rangle_0$. We may note again that the priority rule is completely arbitrary but it facilitates the calculation.

Adopting the priority rule we may now consider the types of vertices for a general problem. We classify the vertex where $L_{\alpha\gamma}$ resides as a diagonal, raising or lowering vertex according to whether α is equal to, greater than, or less than γ. We discuss the three cases separately in the following.

a. Diagonal vertices $L_{\alpha\alpha}$. A diagonal vertex $L_{\alpha\alpha}$ can be an isolated point represented by a small circle and labeled by α. It can also be a center of scattering of Green's functions $G_{\alpha\delta}^0$ or $G_{\gamma\alpha}^0$. The three types of vertices and the associated weight factors are shown in Fig. 4. Note that according to the priority rule all Green's functions $G_{\alpha\delta}^0$ would *always* have the subindices $\alpha > \delta$.

b. Raising vertices $L_{(\alpha+\gamma)\alpha}$. A raising vertex $L_{(\alpha+\gamma)\alpha}$ can create a Green's-function line $G_{(\alpha+\gamma)\alpha}^0$ when it matches with other operators. There is, however, the possibility that it is matched with other raising operators of higher priority; then it will be both a point of termination of a Green's function $G_{\alpha(\alpha-\delta)}^0$ and a point of creation of another Green's function $G_{(\alpha+\gamma)(\alpha-\delta)}^0$. The first vertex can be called a creation vertex and the second one an increment vertex. They are shown in Fig. 5 with the associated weight factors.

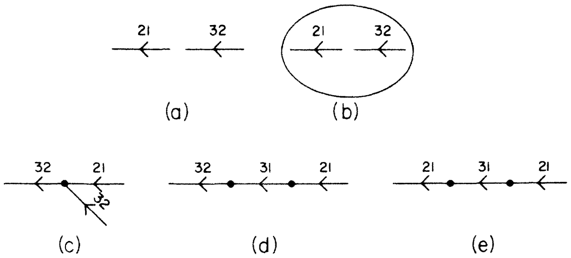

FIG. 2. Diagrams for $\langle T_\tau L_{12}^1(\tau_1)L_{21}^2(\tau_2)L_{32}^3(\tau_3)L_{23}^4(\tau_4)\rangle_0$. Matching begins with $L_{21}^2(\tau_2)$, followed by $L_{32}^3(\tau_3)$ or $L_{31}^3(\tau_3)$. The latter operator is created in the first matching.

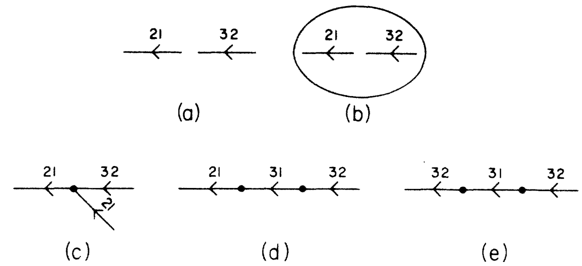

FIG. 3. Diagrams for $\langle T_\tau L_{12}^1(\tau_1)L_{21}^2(\tau_2)L_{32}^3(\tau_3)L_{23}^4(\tau_4)\rangle_0$. Matching begins with $L_{32}^3(\tau_3)$, followed by $L_{21}^2(\tau_2)$ or $L_{31}^3(\tau_2)$. The latter operator is created in the first matching.

c. Lowering vertices $L_{\alpha(\alpha+\gamma)}$. This set of vertices is more complicated than the ones discussed above. We first consider $L_{\alpha(\alpha+1)}$ (i.e., $\gamma = 1$). Being a passive operator, $L_{\alpha(\alpha+1)}$ does not participate in the reduction operation in an active way. It waits for a matching with an active operator. If it is matched with the operator $L_{(\alpha+1)\alpha}$ the vertex becomes the terminating point of the Green's function $G_{(\alpha+1)\alpha}^0$. We call it an annihilation vertex. Noting that as a Green's-function line ends a diagonal vertex is created. Here the vertex carries the operators $L_{\alpha\alpha} - L_{(\alpha+1)(\alpha+1)}$ as a result of matching with $L_{(\alpha+1)\alpha}$, and can again be matched by an active operator of lower or equal priority. This makes the vertex a scattering center. The scattered Green's function, however, would have to have for at least one of its subindices an α or an $\alpha+1$. We have then an annihilation-scattering vertex. Similar to a raising vertex, $L_{\alpha(\alpha+1)}$ can also be both a termination point and a creation point of the Green's functions simultaneously. This is possible if, as a result of matching, the vertex is turned into a raising vertex. There are two general ways of achieving this: (i) matching with $L_{(\alpha+1+\delta)\alpha}$ which turns the vertex into $L_{(\alpha+1+\delta)(\alpha+1)}$, an active operator for $\delta > 0$; (ii) matching with $L_{(\alpha+1)(\alpha-\delta)}$ which turns the vertex into $L_{\alpha(\alpha-\delta)}$, also an active operator. Further matching with the new operator on the vertex follows exactly the discussion of the raising vertices given in (b). The vertices produced are called decrement vertices and are shown along with the other vertices of $L_{\alpha(\alpha+1)}$ in Fig. 6. We note, however, that one type of vertex is missing

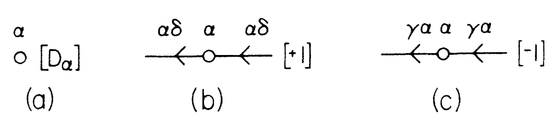

FIG. 4. Diagonal vertices $L_{\alpha\alpha}$ and the associated weight factors (shown in brackets).

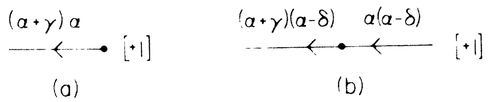

FIG. 5. Raising vertices $L_{(\alpha+\gamma)\alpha}$ and the associated weight factors (shown in brackets).

which could be constructed as a superposition of the increment vertex on the terminal of a Green's function: e.g., a vertex with two incoming Green's-function lines, $G^0_{(\alpha+1+\delta)\alpha}$ and $G^0_{(\alpha+1)\sigma}$, and one outgoing Green's-function line $G^0_{(\alpha+1+\delta)\sigma}$. The priority rule requires that $\alpha+1>\sigma>\alpha$. Therefore it cannot exist. This type of vertex would exist, however, for $\gamma>1$.

In general, for a vertex $L_{\alpha(\alpha+\gamma)}$, an annihilation vertex can be a terminal of one Green's-function line, two Green's-function lines, or as many as γ Green's-function lines. An annihilation-scattering vertex takes the same form as the corresponding annihilation vertex with two additional Green's-function lines (the Green's function being scattered and the scattered Green's function). So there are also γ different types. Within each type, however, there may be more than one kind of vertex. Indeed, for the type of annihilation vertices with ρ lines ($\rho\le\gamma$) merging, there can be $\rho^{(\gamma-\rho)}$ different vertices. They are distinguished by a different set of Green's-function lines which can be arranged as $G^0_{\delta_{i+1}\delta_i}$ ($i=1, 2, \ldots, \rho$), with $\delta_1=\alpha$ and $\delta_{\rho+1}=\alpha+\gamma$. The weight factor for each vertex is $(-1)^{\rho-1}D_{\delta_\rho(\alpha+\gamma)}$.

Similar discussion can be applied to the annihilation-scattering vertices, since such a vertex can be regarded as a superposition of an annihilation vertex and a scattering vertex. The weight factors are, however, different because of the further matching of $L_{\sigma\lambda}$ (of the scattered Green's function) with the diagonal operators $L_{\delta_\rho\delta_\rho}-L_{(\alpha+\gamma)(\alpha+\gamma)}$ now on the annihilation terminal. It should, however, be noted that the priority rule requires $L_{\sigma\lambda}$ be of lower or equal priority than $L_{(\alpha+\gamma)\delta_\rho}$ (i.e., $\lambda\ge\delta_\rho$, $\sigma\ge\alpha+\gamma$). The weight factor for an annihilation-scattering vertex is therefore (i) $(-1)^{\rho-1}$, for $\lambda=\alpha+\gamma$; (ii) $(-1)^\rho$, for $\lambda=\delta_\rho$ and $\sigma>\alpha+\gamma$; (iii) $2(-1)^\rho$, for $\lambda=\delta_\rho$ and $\sigma=\alpha+\gamma$; (iv) 0, otherwise. The two sets of diagrams are shown in Fig. 7(a).

To construct a decrement vertex we need to turn the vertex into a raising vertex first. This may occur after annihilation of one or more Green's-function lines. The ways to activate the vertex are similar to the ones discussed in connection with $L_{\alpha(\alpha+1)}$. After turning the vertex into a raising vertex the instructions given in subsection (b) can be followed. They can be classified into three categories. In the first class, a vertex with ρ incoming Green's-function lines and one outgoing Green's-function line consists of incoming Green's functions $G^0_{\delta_{i+1}\delta_i}$ ($i=1, 2, \ldots, \rho$) with $\delta_1=\alpha$, $\delta_\rho\le\alpha+\gamma-1$ and $\delta_{\rho+1}=\alpha+\gamma^+$ ($\gamma^+>\gamma$), and the outgoing Green's-function line $G^0_{(\alpha+\gamma^+)(\alpha+\gamma)}$. A vertex in the second class can be constructed from the corresponding one in the first class by adding one incoming line $G^0_{(\alpha+\gamma)(\alpha+\gamma^-)}$ and changing the outgoing Green's-function line to $G^0_{(\alpha+\gamma^+)(\alpha+\gamma^-)}$, defining $\gamma^+>\gamma>\gamma^->\delta_\rho$. The last condition is imposed so that the priority rule is observed and thus eliminates the second class of diagrams corresponding to the first class of diagrams with the last incoming Green's-function line as $G_{(\alpha+\gamma^+)(\alpha+\gamma-1)}$. The third class of diagrams has the same incoming Green's-function lines as the corresponding ones in the first class except that the last incoming Green's-function line in each diagram is replaced by $G_{(\alpha+\gamma)\sigma}$, with $\delta_{\rho-1}\le\sigma<\delta_\rho$, and the outgoing Green's-function line is replaced by $G_{\delta_\rho\sigma}$. The weight factor for a decrement vertex with ρ incoming lines is $(-1)^\rho$ for vertices of the first class, $(-1)^{\rho-1}$ for vertices of the second and third classes. The three classes of decrement diagrams are shown in Fig. 7(b).

III. CONSTRUCTION AND EVALUATION OF DIAGRAMS

Diagrams are constructed with the vertices discussed in Sec. II and the interaction lines. In a diagram, all vertices, except the external vertices, are connected by interaction lines. Besides, an oval which is used to restrict the sites of Green's functions and isolated diagonal vertices to a common site can serve also as a means of connecting the Green's functions and the isolated diagonal vertices. Thus a diagram is not merely a set of Green's functions connected by interaction lines, but more generally it is a set of "blocks" connected by interaction lines. A block consists of Green's functions and isolated diagonal vertices as its elements. All elements are restricted to be on a common site. An oval is used to embrace the elements if there are more than one in the block.

As in the fermion or boson cases we draw only topologically distinct diagrams. The evaluation of the diagrams will be done in the Fourier space,

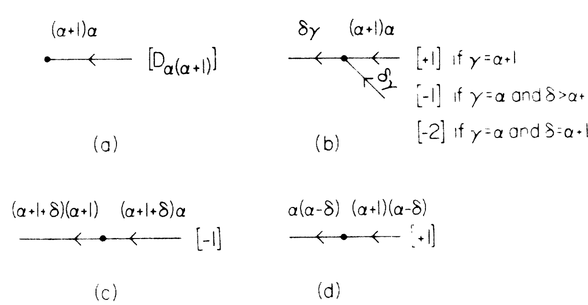

FIG. 6. Lowering vertices $L_{\alpha(\alpha+1)}$ and the associated weight factors (shown in brackets).

as usual, because of the discontinuity of the Green's function at $\tau = 0$. We assume also translational invariance of the system. The Fourier transform of the interaction potential is then

$$V_{\alpha\delta,\gamma\sigma}(q) = \sum_{\vec{r}_m - \vec{r}_n} V^{mn}_{\alpha\delta,\gamma\sigma} e^{i\vec{q}\cdot(\vec{r}_m - \vec{r}_n)} . \tag{22}$$

The Fourier transform of the Green's function is defined as

$$G(\tau) = \sum_n e^{-i\omega_n\tau} G(\omega_n) , \tag{23}$$

$$\omega_n = 2n\pi/\beta , \quad \beta = 1/k_B T ,$$

so

$$G(\omega_n) = \frac{1}{2\beta} \int_{-\beta}^{\beta} G(\tau) e^{i\omega_n\tau} d_\tau , \tag{24}$$

and we obtain

$$G^0_{\alpha\delta}(\omega_n) = 1/\beta(\epsilon_\alpha - \epsilon_\delta - i\omega_n) . \tag{25}$$

The transformation of a diagram from the τr space to the ωq space is a standard procedure. But before we summarize the rules for the calculation of the contribution of a diagram, a discussion of the weight factors for the blocks which contain more than one element will be helpful. The weight factor

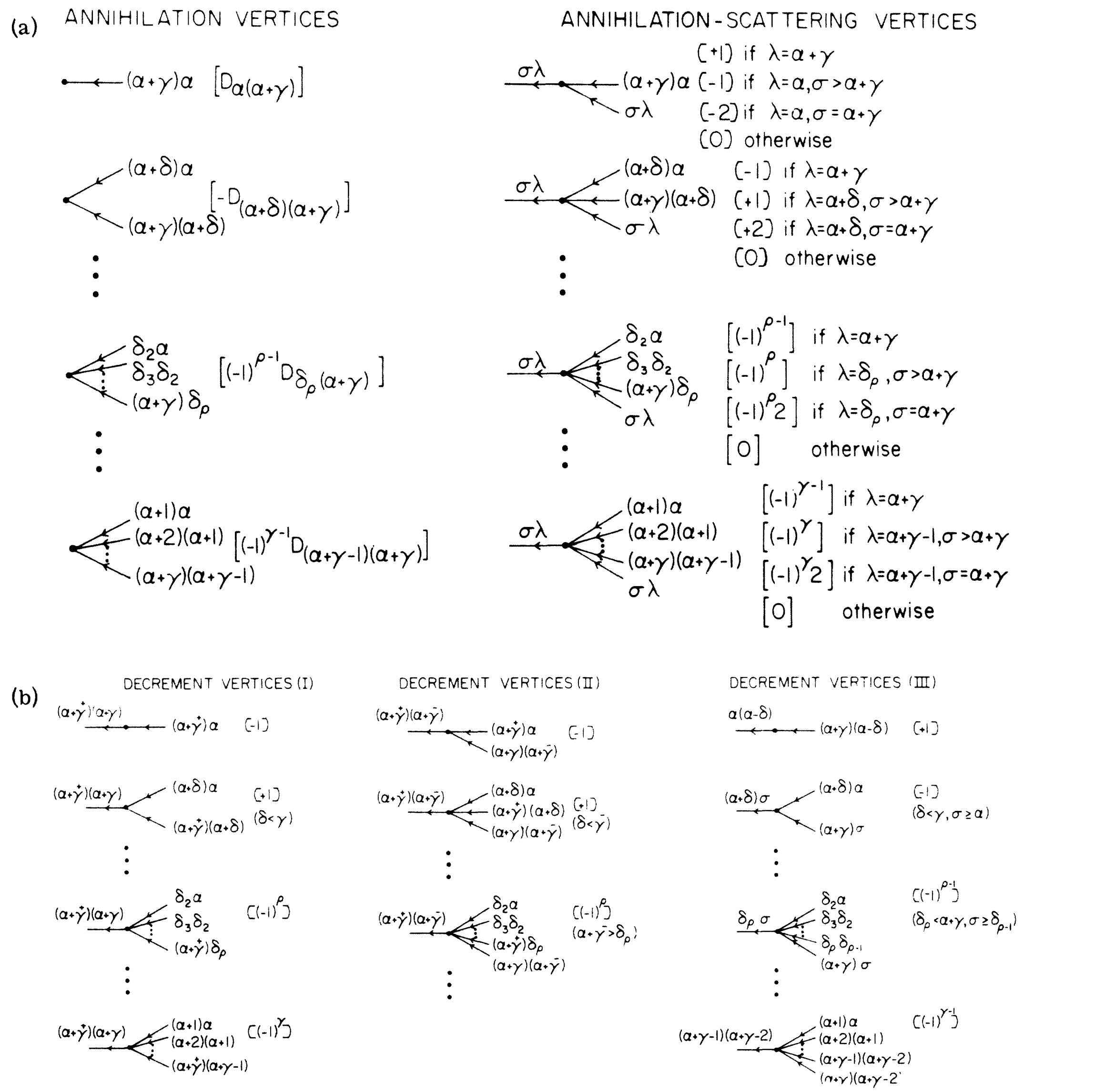

FIG. 7. (a) Lowering vertices $L_{\alpha(\alpha+\gamma)}$ and the associated weight factors (shown in brackets); annihilation vertices and annihilation-scattering vertices. (b) Lowering vertices $L_{\alpha(\alpha+\gamma)}$ and the associated weight factors (shown in brackets); decrement vertices of first, second, and third classes.

carried by each individual vertex has been given previously along with the presentation of the vertices in Figs. 4–7. It was also shown that the weight factor for a block consisting of more than one element is not the product of the weight factors associated with the elements. (The weight factor of the element is clearly the product of the weight factors of the vertices on the element.) To recapitulate the reasoning, we recall that $\langle L_{11}^1 L_{11}^3\rangle_0 = D_1^2 + \delta_{13}(D_1 - D_1^2)$. It was the second term with δ_{13} restricting site 3 to be the same as site 1, that gave rise to the extra diagram with an oval. It is a correction diagram, as discussed previously, to correct the misassignment of the weight factor D_1^2 to the first diagram, in case that site 3 coincides with site 1. To discuss a general case we first find the partition function Z_0 of a single ion in the effective field approximation. We will then show the relation of the weight factors to the derivatives of $\ln Z_0$. The partition function Z_0 is given by

$$Z_0 = \mathrm{Tr}\, e^{-\beta\mathcal{H}_0} = \mathrm{Tr}\exp\left(\sum_\alpha y_\alpha L_{\alpha\alpha}\right), \tag{26}$$

where $y_\alpha = -\beta\epsilon_\alpha$. We first note that

$$D_\alpha = \langle L_{\alpha\alpha}\rangle_0 = \frac{1}{Z_0}\left(\frac{\partial Z_0}{\partial y_\alpha}\right), \tag{27}$$

and

$$\langle (L_{\alpha\alpha})^n\rangle_0 = \frac{1}{Z_0}\left(\frac{\partial^n Z_0}{\partial y_\alpha^n}\right) = \frac{1}{Z_0}\left(\frac{\partial^{n-1}}{\partial y_\alpha^{n-1}}\right)(Z_0 D_\alpha)$$
$$= \sum_m \frac{(n-1)!}{m!\,(n-m-1)!} D_\alpha^{(m)} \langle (L_{\alpha\alpha})^{n-m}\rangle_0 . \tag{28}$$

Some examples of Eq. (28) are

$$\begin{aligned}
&\langle L_{\alpha\alpha}\rangle_0 = D_\alpha ,\\
&\langle (L_{\alpha\alpha})^2\rangle_0 = D_\alpha^2 + D_\alpha' ,\\
&\langle (L_{\alpha\alpha})^3\rangle_0 = D_\alpha^3 + 3D_\alpha D_\alpha' + D_\alpha'' ,\\
&\langle (L_{\alpha\alpha})^4\rangle_0 = D_\alpha^4 + 6D_\alpha^2 D_\alpha' + 3(D_\alpha')^2 + 4D_\alpha D_\alpha'' + D_\alpha''' .
\end{aligned} \tag{29}$$

In fact, since $\langle (L_{\alpha\alpha})^n\rangle_0 = \langle L_{\alpha\alpha}\rangle_0 = D_\alpha$, these equations are nothing but various ways of expressing the same quantity D_α. But with the aid of these expressions we can write down immediately the average of a product of $L_{\alpha\alpha}$'s in a general case:

$$\begin{aligned}
&\langle L_{\alpha\alpha}^1\rangle_0 = D_\alpha ,\\
&\langle L_{\alpha\alpha}^1 L_{\alpha\alpha}^2\rangle_0 = D_\alpha^2 + D_\alpha'\delta_{12} ,\\
&\langle L_{\alpha\alpha}^1 L_{\alpha\alpha}^2 L_{\alpha\alpha}^3\rangle_0 = D_\alpha^3 + D_\alpha D_\alpha'(\delta_{12}+\delta_{23}+\delta_{13}) + D_\alpha''\delta_{12}\delta_{23} ,\\
&\langle L_{\alpha\alpha}^1 L_{\alpha\alpha}^2 L_{\alpha\alpha}^3 L_{\alpha\alpha}^4\rangle_0 = D_\alpha^4 + D_\alpha^2 D_\alpha'(\delta_{12}+\delta_{13}+\delta_{14}+\delta_{23}+\delta_{24}+\delta_{34}) + (D_\alpha')^2(\delta_{12}\delta_{34}+\delta_{13}\delta_{24}+\delta_{14}\delta_{23}) + D_\alpha'' D_\alpha\\
&\qquad\times(\delta_{12}\delta_{23}+\delta_{12}\delta_{14}+\delta_{13}\delta_{34}+\delta_{23}\delta_{34}) + D_\alpha'''\delta_{12}\delta_{23}\delta_{34} .
\end{aligned} \tag{30}$$

It is clear that for a block consisting of $nL_{\alpha\alpha}$ vertices, the weight factor should be $D_\alpha^{(n-1)}$. Since $D_\alpha = \partial_\alpha \ln Z_0$, the first derivative of $\ln Z_0$ with respect to y_α, the weight factor can also be written as $\partial_\alpha^n \ln Z_0$.

It is a simple matter to extend the above argument to a more general case. A block containing $n_\alpha L_{\alpha\alpha}$, $n_\delta L_{\delta\delta}$, etc., would be given a weight factor $\partial_\alpha^{n_\alpha}\partial_\delta^{n_\delta}\cdots\partial_\gamma^{n_\gamma}\ln Z_0$. We now turn our attention to consider the Green's-function lines which may be embraced in an oval. Certain vertices of Green's-function lines carry weight factors like $D_{\alpha\delta}$. If a block contains $n_{\alpha\delta}$ such vertices, the weight factor is $\partial_{\alpha\delta}^{n_{\alpha\delta}}\ln Z_0$, where $\partial_{\alpha\delta} \equiv \partial_\alpha - \partial_\delta$. In the most general case where a block contains n_α vertices of weight factor D_α, $n_{\alpha\delta}$ vertices of weight factor $D_{\alpha\delta}$, etc., the weight factor should be $\partial_\alpha^{n_\alpha}\cdots\partial_\gamma^{n_\gamma}\partial_{\alpha\delta}^{n_{\alpha\delta}}\cdots\partial_{\gamma\sigma}^{n_{\gamma\sigma}}\ln Z_0$.

The contribution in ωq space of a diagram from the expansion of the Green's function can now be obtained in the following way.

(a) Assign a frequency to each Green's-function line and a momentum to each interaction line. The frequencies are conserved at each vertex and the momenta are conserved on each block.

(b) Associate with each Green's function the appropriate factor, $G_{\alpha\delta}^0(\omega_n) = 1/\beta(\epsilon_\alpha - \epsilon_\delta - i\omega_n)$.

(c) Associate with the interaction line connecting vertices $L_{\alpha\delta}$ and $L_{\gamma\sigma}$, a factor $-\beta V_{\alpha\delta,\gamma\sigma}(q)$.

(d) Associate with each block the corresponding weight factor; a weight factor $(-1)^{n_m}2^{n_t}\partial_\alpha^{n_\alpha}\cdots\partial_\gamma^{n_\gamma}\partial_{\alpha\delta}^{n_{\alpha\delta}}\cdots\partial_{\gamma\sigma}^{n_{\gamma\sigma}}(\ln Z_0)$ is assigned to a block containing n_m vertices carrying a minus sign, n_t vertices carrying a factor 2 or -2, and n_α vertices of weight factor D_α, $n_{\alpha\delta}$ verrices of weight $D_{\alpha\delta}$, etc.

(e) Multiply all the factors from (b) to (d) and divide it by the symmetry factor of the diagram. The symmetry factor is equal to the number of ways in which the diagram can be rotated and reflected into itself. Finally we should integrate over all internal frequencies and sum over all internal momenta.

IV. SUMMATION OF DIAGRAMS; A SIMPLE EXAMPLE

Before discussing the summation of diagrams, we note the the linked-cluster theorem applies in the usual way. Thus we can discard all the disconnected diagrams. As we recall, Dyson's equation is commonly used in summing diagrams in the fermion and boson systems. Complication arises in a spin system either in the formalism using spin operators as done by VLP, or using the standard basis operators presented here. The main obstacle for using Dyson's equation in its original form is the fact that the weight factor for the part of a diagram contained in an oval is not the product of the weight factors of the constituents considered separately. We must therefore modify Dyson's equation. Following VLP we define an irreducible part as a diagram, or part of a diagram, which would not break apart by removing one interaction line. Then a diagram can be viewed as a set of irreducible parts joined together by single interaction lines. We may collect all the irreducible parts and denote the collection Σ. The modified Dyson equation can be represented symbolically as,

$$G = \Sigma + \Sigma V G \ . \tag{31}$$

Thus the calculation of a Green's function is now reduced to the calculation of the irreducible parts. Generally it is impossible to sum over all the diagrams. Classification of the diagrams according to certain physical parameters is essential in the selection of diagrams for the summation.

In a system of localized magnetic ions the reciprocal of the effective number of ions interacting with a given ion (or the reciprocal of the range of interaction) has been used as the expansion parameter. The leading order diagrams are the chain-like diagrams. We illustrate here the use of the modified Dyson's summation formula to sum over the chain diagrams. The simplicity in this approximation, however, precludes a full demonstration of the method. A more general use of the formula will be given in a subsequent calculation of an anisotropic ferromagnetic system.

We calculate the Green's function $G_{\alpha\delta,\delta\alpha}(\omega, q)$, the ωq Fourier component of $\langle T_\tau(L^m_{\alpha\delta}(\tau)L^n_{\delta\alpha}(0))\rangle$. To simplify the notations we shall use a roman letter to denote the two Greek letters; i stands for $\alpha\delta$, i^+ for $\delta\alpha$, and so on. The only irreducible part in this approximation is the single Green's function lines, provided that $\omega_n \neq 0$ or there are no diagonal interaction vertices. We obtain

$$G_{ji^+}(\omega_n, q) = g_{ii^+}(\omega_n)\delta_{ji} + \sum_l g_{jj^+}(\omega_n) V^{j^+l} G_{li^+}(\omega_n, q) \ , \tag{32}$$

where

$$g_{ii^+}(\omega_n) = \frac{D_{\alpha\delta}}{\beta(\epsilon_\delta - \epsilon_\alpha - i\omega_n)} \tag{33}$$

and

$$V^{i^+j}(q) = -\beta V_{\delta\alpha,\gamma\sigma}(q) \ . \tag{34}$$

Let G and g be matrices with elements G_{ij} and g_{ii^+} respectively. Equation (32) can then be solved in the matrix form as

$$G = (g^{-1} - V)^{-1} \ . \tag{35}$$

If we further assume a simple product form for the potential of interaction,

$$V_{\delta\alpha,\gamma\sigma}(q) = V(q) C^*_{\alpha\delta} C_{\gamma\sigma} \ , \tag{36}$$

which is often true, we can obtain for G_{ji^+} in a simple form. We first define

$$\tilde{g}_{ii^+} = C_i g_{ii^+} C^*_i \tag{37}$$

and

$$\tilde{G}_{ji^+} = C_j G_{ji^+} C^*_i \ ; \tag{38}$$

we find

$$\tilde{G}_{ji^+} = \frac{V(q)\tilde{g}_{jj^+}\tilde{g}_{ii^+} + \delta_{ij}\tilde{g}_{ii^+}[1 - V(q)\sum_l \tilde{g}_{ll^+}]}{1 - V(q)\sum_l \tilde{g}_{ll^+}} \ , \tag{39}$$

or

$$G_{ji^+} = \frac{V(q)C^*_j C_i g_{jj^+} g_{ii^+} + \delta_{ij} g_{ii^+}[1 - V(q)\sum_l C^*_l C_l g_{ll^+}]}{1 - V(q)\sum_l g_{ll^+} C^*_l C_l} \ . \tag{40}$$

Equation (40) gives the various Green's functions in the chain-diagram approximation (zeroth order in $1/z$). Physical susceptibility functions can be constructed by taking proper combinations of the Green's functions. In particular, the susceptibility function corresponding to the field and moment involved in the interaction potential is given by the sum of the weighted Green's functions of Eq. (39),

$$\sum_{ij} \tilde{G}_{ji^+} = \frac{\sum_l \tilde{g}_{ll^+}}{1 - V(q)\sum_l \tilde{g}_{ll^+}} \ . \tag{41}$$

This equation reduces to the one obtained by Fulde and Peschel[11] for a two-level induced-moment system and the one obtained by Holden and Buyers[12] in the discussion of the induced-moment system Pr_3Tl.

In conclusion, we emphasize again the versatility of the present method. Using the technique derived above the calculation of collective excitations such as spin waves and magnetic excitons, for systems with large crystal-field anisotropy, or in the presence of a quadrupole-quadrupole interaction or higher-order multipole-multipole interactions, is a simple matter. Interactions of the excitations can also be considered. The method is indispensable in dealing with rare-earth and actinide systems. It not only provides a systematic and consistent approach for treating the ion-ion interaction, but also can be used in discussing the interactions of ions with other particles in the system, such as phonons and electrons. Calculations employing the present method on the anisotropic ferromagnetic system and a magnetic system with both bilinear and biquadratic exchange interactions are underway.

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